

```
clc;

clear;

close all;

% Specifications

Rp = 1;    % Passband Ripple (dB)

Rs = 40;   % Stopband Attenuation (dB)

Wp = 0.5;  % Passband Edge

Ws = 0.35; % Stopband Edge

% Find Filter Order

[n,Wn] = cheb1ord(Wp,Ws,Rp,Rs);

% Design High-Pass Filter

[b,a] = cheby1(n,Rp,Wn,'high');

% Frequency Response

figure;

freqz(b,a);

title('Frequency Response');

% Impulse Response

figure;

impz(b,a);

title('Impulse Response');

% Group Delay

figure;

grpdelay(b,a);

title('Group Delay');

% Pole-Zero Plot

figure;

zplane(b,a);

title('Pole-Zero Plot');
```

```
fprintf('Filter Order = %d\n',n);  
fprintf('Cutoff Frequency = %.4f\n',Wn);
```



